Dual-Task Interference

Modality & Processing Code Effects on Primary Task Performance

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Introduction & Motivation

Multiple Resource Theory (MRT) predicts that interference is greatest when tasks share the same perceptual modality and processing code. This study tests that prediction in a controlled dual-task experiment to identify which resource combinations most degrade performance and workload.

Theoretical Framework

- **Multiple Resource Theory**: interference increases when tasks share attentional channels - modality and/or processing code
- **Modality**: Visual vs. Auditory
- **Processing Code**: Spatial vs. Verbal
- **Key Prediction**: Cross-modal, cross-code pairings produce less interference than same-modality, same-code pairings

Methodology

- **Design**: 2x2 within subjects; randomized counterbalancing, n = 40
- **Primary Task**: Continuous visual-spatial cursor tracking (25s trials); performance measured as mean Euclidean distance from target
- **Secondary Tasks**: Visual-Spatial (VS), Visual-Verbal (VV), Auditory-Spatial (AS), Auditory-Verbal (AV); probes delivered every 3-5s
- **Workload Assessment**: Partial NASA-TLX after each condition
- **Analysis**: Two-way repeated measures ANOVA (modality x code), interaction plots

Explore the full project on GitHub!

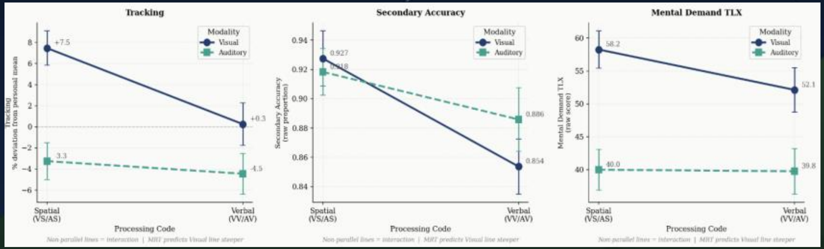

Figure 1: Tracking/Secondary/Mental Demand Interaction Plot

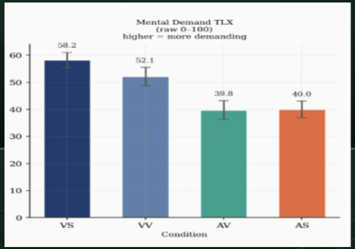

Figure 2: Mental Demand NASA TLX Means

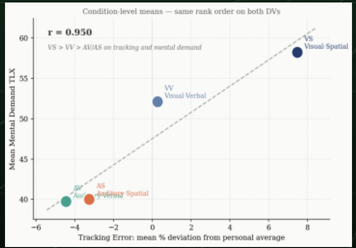

Figure 3: Workload-Performance Alignment Plot

Figure 4: Primary Task Experiment Design

Figure 5: VS-VS Experiment Condition

Results

- **Tracking Error**: Modality significant ($F = 8.78$, $p = 0.005$); visual secondary tasks caused greater disruption
- **Secondary Accuracy**: Code significant ($F = 5.94$, $p = 0.020$); verbal tasks answered less accurately
- **Mental Demand**: Modality significant ($F = 28.08$, $p < 0.001$); visual conditions rated more demanding
- **No significant interaction effects** were found for any dependent variable
- **Rank order**: VS > VV > AS > AV, consistent with MRT predictions, tracking error and mental demand captured near-perfect correlation across conditions with $r = 0.95$

Discussion & Conclusions

- Results partially support MRT; **modality** drove tracking error and workload, while **processing code** drove secondary accuracy, suggesting separable resource pools
- Visual secondary tasks impose a disproportionate cost on a VS primary task, consistent with within-modality competition
- Alignment between tracking performance and mental demand TLX strengthens both measures' validities

Next Steps

- Standardize testing environment and hardware across administrators
- Expand to all four MRT axes
- Increase sample size for greater statistical power
- Test across varied primary task types
- Enforce consistency in task type (discrete vs. continuous) across primary and secondary tasks
- Explore alternate primary task metrics to account for participant strategy